امتحان عملي خوارزميات البحث الذكية: الاسم الثلاثي: السنة:
المدة: 30 دقيقة الرقم الجامعي:

1- Does the following code represent a pseudo code for A* if no , correct the mistakes in the code

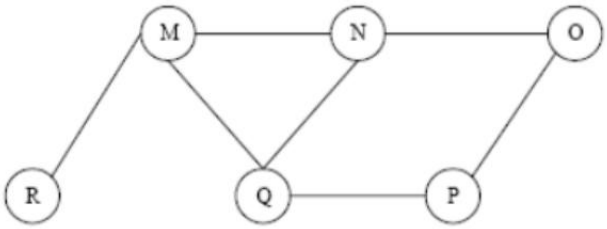
```
frontier ← a queue
visited ← a queue
initialNode ← new Node(problem.getInitialNode())
frontier.push(initialNode)
while not isEmpty(frontier) do
    currentNode ← pull(frontier)
    if (currentNode.getState.isGoal())
        return currentNode
    for (action a in problem.actions)
        children ← a.apply for currentNode.state
            for (child in children)
                childNode ← new Node()
                if (childNode.getState.isGoal())
                    return childNode
                if (!visited.contain(child)
                        frontier.push(childNode)
return fail
```

2- if you want to solve N-Queens problem and you are not interested with the path, you need to find a good solution with good time and space what do you use:
a- A* b- Hill climbing

3- which of the algorithms fails to find shortest path in weighted graph
a- BFS b-DFS c- A*

4- consider the following problem: we have a chess board 8*8, we have a knight on this board placed on certain position (for example 5*2) and we want to move it to another position (for example 3*7) we want to find the shortest steps for that, suggest an algorithm can be used for this purpose

5- for this graph



One of the possible orders of visiting the graph in BFS:

a- MNQPQR
b-NQMPOR
c-QMNPRO
d-QMNPOR

6- A vacuum cleaner robot operates in a house consisting of two rooms, A and B. Each room can be either clean or dirty. The robot starts in one of the rooms, and its task is to clean both rooms.
a- Suggest a suitable state representation.
b- Suggest a goal test.
c- Suggest the possible actions.

7- Write a pseudo code for the hill climbing where the we need minimum optimal